

Practical 13 - Alternative models for survival data

Survival Analysis

London School of Hygiene & Tropical Medicine

SOLUTIONS

Datasets required: `pbcbase.csv`

R packages required: `survival`, `flexsurv`, `mexhaz`, `timereg`

```
library(survival)
library(flexsurv)
library(mexhaz)
library(timereg)
```

Introduction

In this practical we will return to using the `pbc` data, used in earlier practicals. The data are from a randomised trial and the time origin is the time of randomisation. The variables of interest for this practical are described below.

Variable	Description
<code>id</code>	Unique identifier for each participant
<code>d</code>	Event indicator at the end of follow-up: 0=alive (censored), 1=dead
<code>time</code>	Follow-up time in years
<code>treat</code>	Treatment: 1=placebo, 2=active
<code>bil0</code>	Serum bilirubin (mg/dl), measured at the start of the trial.

The practical is in 3 parts:

- Part A: Flexible parametric models
- Part B: Accelerated failure time models
- Part C: Additive hazards models

Load the data using:

```
pbcbase<-read.table("datasets/pbcbase.csv",sep=" ",header=T)
```

Aims

The aims of this session are:

- Fit flexible parametric models (FPM) using a spline for the baseline hazard, and obtain corresponding survival curves.
- Fit Weibull models and make the link between proportional hazards and accelerated failure time (AFT) parametrizations.
- Fit Weibull and log-logistic AFT models and compare the fit.
- Fit additive hazard models and interpret the graphical output.

Part A: Flexible parametric models

Question 1

In this question we will fit a flexible parametric model (FPM) to the PBC data, with the baseline hazard modelled using a spline (as a function of time).

Use the code below to carry out the following steps: - Start by calculating the 25th, 50th and 75th percentiles (i.e. the quartiles) of the log follow-up time for individuals who have the event. We call these `myknots`. - Using `flexsurvspline` fit a FPM including `treat` and `bil0` as covariates. For the baseline hazard use a natural cubic spline with 3 interior knots for $\log(t)$ located at the quartiles of log follow-up time in people who have the event. This can be done by specifying `knots=myknots` and `timescale="log"` (this is the default for `timescale`). - The same thing can be done using `flexsurvspline` and simply specifying the number of knots as `k=3`, which will place 3 interior knots at the quartiles of $\log(t)$ in people who have the event. - Take a look at what the output is showing and link this to what was shown in the slides.

```
#obtain quartiles of log(t)
myknots<-quantile(log(pbc$time[pbc$d==1]),probs=c(0.25,0.5,0.75))

#fit FPM using above knot placements
```

```

pbc.flex1<-flexsurvspline(Surv(time,d)~treat+bil0,data=pbc,
                          knots=myknots,timescale="log",spline="rp")
pbc.flex1

#same thing using k=3
pbc.flex2<-flexsurvspline(Surv(time,d)~treat+bil0,data=pbc,
                          k=3,timescale="log")
pbc.flex2

```

SOLUTION

```

#obtain quartiles of log(t)
myknots<-quantile(log(pbc$time[pbc$d==1]),probs=c(0.25,0.5,0.75))

#fit FPM using above knot placements
pbc.flex1<-flexsurvspline(Surv(time,d)~treat+bil0,data=pbc,
                          knots=myknots,timescale="log",spline="rp")
pbc.flex1

```

Call:

```

flexsurvspline(formula = Surv(time, d) ~ treat + bil0, data = pbc,
               knots = myknots, timescale = "log", spline = "rp")

```

Estimates:

	data	mean	est	L95%	U95%	se	exp(est)
gamma0		NA	-2.410994	-3.613891	-1.208097	0.613734	NA
gamma1		NA	0.782741	0.192212	1.373270	0.301296	NA
gamma2		NA	0.055762	-0.298233	0.409758	0.180613	NA
gamma3		NA	-0.021477	-1.286347	1.243393	0.645354	NA
gamma4		NA	-0.181478	-1.649506	1.286550	0.749008	NA
treat	1.510870		-0.419172	-0.841373	0.003028	0.215412	0.657591
bil0	61.431993		0.007125	0.005998	0.008252	0.000575	1.007151
	L95%		U95%				
gamma0		NA		NA			
gamma1		NA		NA			
gamma2		NA		NA			
gamma3		NA		NA			
gamma4		NA		NA			
treat	0.431118		1.003033				

```
bil0      1.006016    1.008287
```

```
N = 184,  Events: 96,  Censored: 88  
Total time at risk: 747.039  
Log-likelihood = -267.7807, df = 7  
AIC = 549.5614
```

```
#same thing using k=3  
pbc.flex2<-flexsurvspline(Surv(time,d)~treat+bil0,data=pbc,  
                           k=3,timescale="log")  
pbc.flex2
```

Call:

```
flexsurvspline(formula = Surv(time, d) ~ treat + bil0, data = pbc,  
               k = 3, timescale = "log")
```

Estimates:

	data	mean	est	L95%	U95%	se	exp(est)
gamma0		NA	-2.410994	-3.613891	-1.208097	0.613734	NA
gamma1		NA	0.782741	0.192212	1.373270	0.301296	NA
gamma2		NA	0.055762	-0.298233	0.409758	0.180613	NA
gamma3		NA	-0.021477	-1.286347	1.243393	0.645354	NA
gamma4		NA	-0.181478	-1.649506	1.286550	0.749008	NA
treat	1.510870		-0.419172	-0.841373	0.003028	0.215412	0.657591
bil0	61.431993		0.007125	0.005998	0.008252	0.000575	1.007151
	L95%		U95%				
gamma0		NA		NA			
gamma1		NA		NA			
gamma2		NA		NA			
gamma3		NA		NA			
gamma4		NA		NA			
treat	0.431118		1.003033				
bil0	1.006016		1.008287				

```
N = 184,  Events: 96,  Censored: 88  
Total time at risk: 747.039  
Log-likelihood = -267.7807, df = 7  
AIC = 549.5614
```

- The hazard ratio for treatment (active vs placebo) is 0.66 (95% CI 0.43 to 1.00), conditional on bilirubin. Given bilirubin, there is some evidence that active treatment is associated with a lower hazard. The hazard ratio for bilirubin is 1.007 (95% CI 1.006,

1.008), conditional on treatment. That is, for a 1 unit increase in bilirubin the hazard increases by 0.7%, given treatment, and there is strong evidence of an association.

- The other parameter estimates shown in the output are the coefficients for the spline terms that make up the baseline hazard. These do not individually have a useful interpretation, but in combination they can be used to obtain the shape of the baseline hazard, and hence other quantities such as a survival probability.

Question 2

- Use the code below to obtain estimated survival curves based on the FPM fitted in question 1 for two individuals: person 1 has `treat=1`, `bil0=30` and person 2 has `treat=2`, `bil0=30`.
- Fit a new FPM with 10 internal knots instead of 3. Compare the coefficients from the two models.
- Compare estimated survival curves from the two models for the two individuals above.

```
#Obtain estimated survival; probabilities at times 0,0.1,0.2,...,11.9,12
surv.flex1<-predict(pbc.flex1,
                    newdata=data.frame(treat=c(1,2),bil0=c(30,30)),
                    type="survival",times=seq(0,12,0.1))

#Plot survival curves
times<-seq(0,12,0.1)
surv.person1.flex1<-surv.flex1$.pred[[1]]$.pred_survival
surv.person2.flex1<-surv.flex1$.pred[[2]]$.pred_survival

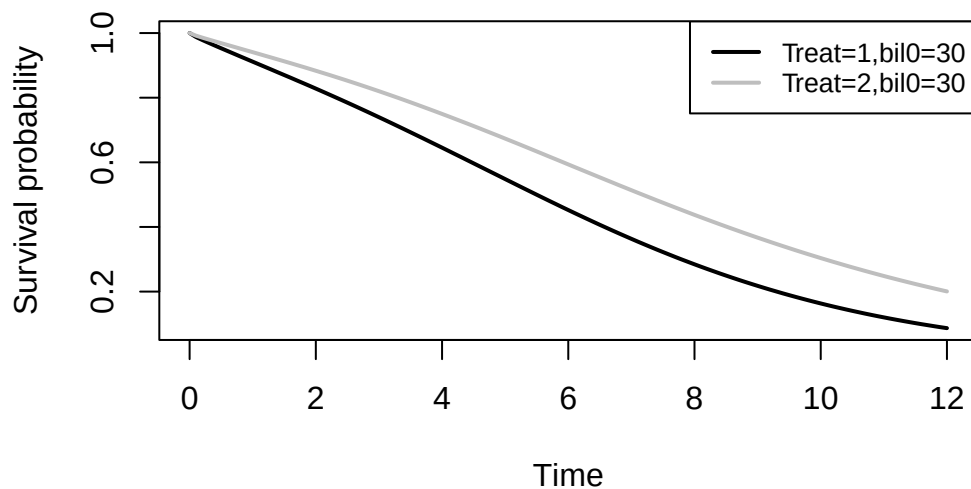
plot(times, surv.person1.flex1, type="l",lwd=2,
      xlab="Time",ylab="Survival probability")
lines(times, surv.person2.flex1, type="l",lwd=2,col="grey")
legend(x="topright",c("Treat=1,bil0=30","Treat=2,bil0=30"),
      lty=1,lwd=2,col=c("black","grey"),cex=0.8)
```

SOLUTION

```
#Obtain estimated survival; probabilities at times 0,0.1,0.2,...,11.9,12
surv.flex1<-predict(pbc.flex1,
                    newdata=data.frame(treat=c(1,2),bil0=c(30,30)),
                    type="survival",times=seq(0,12,0.1))
```

```
#Plot survival curves
times<-seq(0,12,0.1)
surv.person1.flex1<-surv.flex1$.pred[[1]]$.pred_survival
surv.person2.flex1<-surv.flex1$.pred[[2]]$.pred_survival

plot(times, surv.person1.flex1, type="l",lwd=2,
      xlab="Time",ylab="Survival probability")
lines(times, surv.person2.flex1, type="l",lwd=2,col="grey")
legend(x="topright",c("Treat=1,bil0=30","Treat=2,bil0=30"),
      lty=1,lwd=2,col=c("black","grey"),cex=0.8)
```



```
#Fit model with 10 interior knots
pbc.flex10<-flexsurvspline(Surv(time,d)~treat+bil0,data=pbc,
                           k=10,timescale="log",spline="rp")
pbc.flex10
```

Call:

```
flexsurvspline(formula = Surv(time, d) ~ treat + bil0, data = pbc,
               k = 10, timescale = "log", spline = "rp")
```

Estimates:

	data	mean	est	L95%	U95%	se	exp(est)
gamma0	NA		-2.15377	-3.96856	-0.33899	0.92593	NA
gamma1	NA		0.87098	0.10194	1.64003	0.39238	NA
gamma2	NA		-0.01077	-0.62440	0.60286	0.31308	NA
gamma3	NA		2.10581	-2.68483	6.89645	2.44425	NA
gamma4	NA		-8.42285	-21.08936	4.24367	6.46263	NA

gamma5	NA	10.49673	-2.68225	23.67571	6.72409	NA
gamma6	NA	-9.48105	-25.56633	6.60423	8.20693	NA
gamma7	NA	10.76312	-17.23161	38.75785	14.28329	NA
gamma8	NA	-11.33179	-60.24221	37.57862	24.95475	NA
gamma9	NA	8.78760	-62.77947	80.35468	36.51449	NA
gamma10	NA	-5.43698	-60.13385	49.25989	27.90708	NA
gamma11	NA	3.76392	-16.93554	24.46338	10.56115	NA
treat	1.51087	-0.41087	-0.83290	0.01116	0.21533	0.66308
bil0	61.43199	0.00714	-0.00283	0.01711	0.00509	1.00716
	L95%		U95%			
gamma0	NA		NA			
gamma1	NA		NA			
gamma2	NA		NA			
gamma3	NA		NA			
gamma4	NA		NA			
gamma5	NA		NA			
gamma6	NA		NA			
gamma7	NA		NA			
gamma8	NA		NA			
gamma9	NA		NA			
gamma10	NA		NA			
gamma11	NA		NA			
treat	0.43479		1.01123			
bil0	0.99717		1.01726			

N = 184, Events: 96, Censored: 88
 Total time at risk: 747.039
 Log-likelihood = -266.1519, df = 14
 AIC = 560.3039

```

#Obtain estimated survival; probabilities at times 0,0.1,0.2,...,11.9,12
surv.flex10<-predict(pbc.flex10,
                     newdata=data.frame(treat=c(1,2),bil0=c(30,30)),
                     type="survival",times=seq(0,12,0.1))

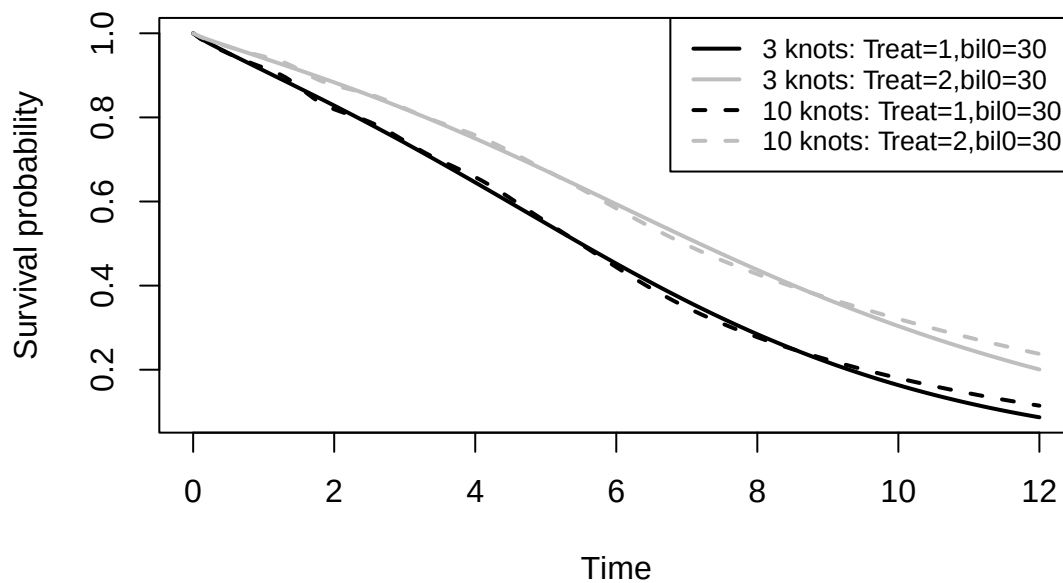
#Plot survival curves, and compare with those from above
surv.person1.flex10<-surv.flex10$.pred[[1]]$.pred_survival
surv.person2.flex10<-surv.flex10$.pred[[2]]$.pred_survival

plot(times, surv.person1.flex1, type="l",lwd=2,
      xlab="Time",ylab="Survival probability")
lines(times, surv.person2.flex1, type="l",lwd=2,col="grey")
  
```

```

lines(times, surv.person1.flex10, type="l",lwd=2,col="black",lty=2)
lines(times, surv.person2.flex10, type="l",lwd=2,col="grey",lty=2)
legend(x="topright",
      c("3 knots: Treat=1,bil0=30","3 knots: Treat=2,bil0=30",
        "10 knots: Treat=1,bil0=30","10 knots: Treat=2,bil0=30"),
      lty=c(1,1,2,2),lwd=2,
      col=c("black","grey","black","grey"),cex=0.8)

```



- Using 10 knots results in a survival curve with more 'bends'. It is more flexible, but we should take care about overfitting.

Question 3

- Compare your hazard ratio estimates from the FPM (with 3 knots) with those from using a Cox proportional hazards model.
- Compare the estimated survival curves for the two individuals considered above.
- What are the assumptions of the FPM versus the Cox model?

SOLUTION

```
#Fit Cox PH model
pbc.cox<-coxph(Surv(time,d)~as.factor(treat)+bil0,data=pbc)
summary(pbc.cox)
```

Call:

```
coxph(formula = Surv(time, d) ~ as.factor(treat) + bil0, data = pbc)
```

n= 184, number of events= 96

	coef	exp(coef)	se(coef)	z	Pr(> z)
as.factor(treat)2	-0.3838118	0.6812596	0.2142518	-1.791	0.0732 .
bil0	0.0071108	1.0071361	0.0008348	8.518	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
as.factor(treat)2	0.6813	1.4679	0.4477	1.037
bil0	1.0071	0.9929	1.0055	1.009

Concordance= 0.746 (se = 0.026)

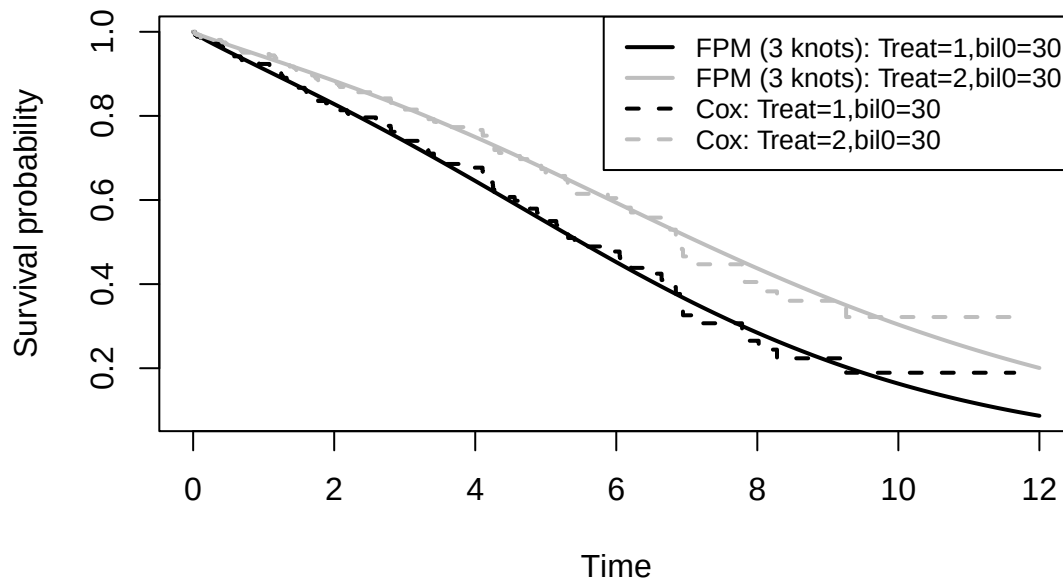
Likelihood ratio test= 46.6 on 2 df, p=8e-11

Wald test = 72.98 on 2 df, p=<2e-16

Score (logrank) test = 96.22 on 2 df, p=<2e-16

```
#Obtain survival probabilities
surv.cox<-survfit(pbc.cox,
                  newdata=data.frame(treat=c(1,2),bil0=c(30,30)))
surv.person1.cox<-surv.cox$surv[,1]
surv.person2.cox<-surv.cox$surv[,2]

#Plot survival curves, and compare with those from above
plot(times, surv.person1.flex1, type="l",lwd=2,
      xlab="Time",ylab="Survival probability")
lines(times, surv.person2.flex1, type="l",lwd=2,col="grey")
lines(surv.cox$time, surv.person1.cox, type="s",lwd=2,col="black",lty=2)
lines(surv.cox$time, surv.person2.cox, type="s",lwd=2,col="grey",lty=2)
legend(x="topright",
      c("FPM (3 knots): Treat=1,bil0=30","FPM (3 knots): Treat=2,bil0=30",
        "Cox: Treat=1,bil0=30","Cox: Treat=2,bil0=30"),
      lty=c(1,1,2,2),lwd=2,
      col=c("black","grey","black","grey"),cex=0.8)
```



- The HR estimate for treatment is 0.66 (95% CI 0.43 to 1.00) from the FPM and 0.68 (95% CI 0.45 to 1.04) from the Cox model. The HR estimate for bilirubin is 1.007 (95% CI 1.006, 1.008) from the FPM and 1.007 (95% CI 1.006 to 1.009) from the Cox model. That is, the estimates are almost identical.
- The estimated survival curves based on the FPM are smooth, whereas the curves based on the Cox model are formed from steps. However, the two models give generally very similar curves.

Part B: Accelerated failure time models

Question 1

- Use `mexhaz` to fit a Weibull model with treatment group as the explanatory variable, using the proportional hazards parametrization.
- Write down the model being fitted and interpret your results.

```
#Weibull model using PH model
mod.weib.ph=mexhaz(Surv(time,d)~as.factor(treat),data=pbpc,
                   base="weibull")
summary(mod.weib.ph)
```

SOLUTION

```
#Weibull model using PH model
mod.weib.ph=mexhaz(Surv(time,d)~as.factor(treat),data=pbcb,
                    base="weibull")
```

```
iteration = 0
Step:
[1] 0 0 0
Parameter:
[1] -1.025885  0.000000  0.000000
Function Value
[1] 366.2829
Gradient:
[1] 171.79796 276.08505  93.53759

iteration = 5
Parameter:
[1] -2.07399771  0.05721056 -0.14810807
Function Value
[1] 292.5257
Gradient:
[1] 4.143445e-09 3.461338e-09 2.814808e-09
```

Relative gradient close to zero.
Current iterate is probably solution.

Data

Name	N.Obs.Tot	N.Obs	N.Events	N.Clust
pbcb	184	184	96	1

Details

Iter	Eval	Base	Nb.Leg	Nb.Aghq	Optim	Method	Code	LogLik	Total.Time
5	8	weibull	20	10	nlm	---	1	-292.5257	0.01

```
summary(mod.weib.ph)
```

Call:
mexhaz(formula = Surv(time, d) ~ as.factor(treat), data = pbcb,

```

base = "weibull")

Coefficients:
              Estimate      StdErr   z.value p.value
logLambda      -2.073998    0.206510 -10.0431  <2e-16 ***
logRho           0.057211    0.088060   0.6497   0.5159
as.factor(treat)2 -0.148108    0.204500  -0.7242   0.4689
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Hazard ratios (for proportional effect variables):
              Coef      HR CI.lower CI.upper
as.factor(treat)2 -0.1481 0.8623   0.5776   1.2875

log-likelihood: -292.5257 (for 3 degree(s) of freedom)

number of observations: 184, number of events: 96

```

- The form of the hazard model being fitted is $h(t|X) = \kappa \lambda t^{\kappa-1} e^{\beta_{PH} X}$, where X denotes treatment (Active: $X = 1$, Control: $X = 0$).
- The log hazard ratio is $\hat{\beta}_{PH} = -0.148$. The estimate of κ is $e^{0.057211} = 1.059$ (looking at `logRho`).
- The hazard ratio estimate is 0.86. The interpretation is that, according to the point estimate, the hazard is lower by 14% in the active treatment group versus in the control group. The 95% CI is (0.58, 1.29), so there is no evidence here against the null hypothesis that treatment is associated with a reduction in the hazard for mortality.

Question 2

- Use `survreg` to fit a Weibull model with treatment group as the explanatory variable, using the AFT parametrization. Note that `survreg` uses the AFT parametrization for the Weibull model by default.
- Write down the form of the survival function under the AFT Weibull model and interpret your results.
- What is the estimate of the acceleration factor $e^{\beta_{AFT}}$? Interpret the estimate.
- What is the relationship between your estimates of the hazard ratio $e^{\beta_{PH}}$ and the acceleration factor $e^{\beta_{AFT}}$?

```

#Weibull model using AFT model
mod.weib.aft=survreg(Surv(time,d)~as.factor(treat),data=abc,
                    dist="weibull")
summary(mod.weib.aft)

```

SOLUTION

```
#Weibull model using AFT model
mod.weib.aft=survreg(Surv(time,d)~as.factor(treat),data=pbpc,dist="weibull")
summary(mod.weib.aft)
```

Call:

```
survreg(formula = Surv(time, d) ~ as.factor(treat), data = pbpc,
        dist = "weibull")
```

	Value	Std. Error	z	p
(Intercept)	1.9587	0.1386	14.13	<2e-16
as.factor(treat)2	0.1399	0.1928	0.73	0.47
Log(scale)	-0.0572	0.0881	-0.65	0.52

Scale= 0.944

Weibull distribution

Loglik(model)= -292.5 Loglik(intercept only)= -292.8

Chisq= 0.52 on 1 degrees of freedom, p= 0.47

Number of Newton-Raphson Iterations: 5

n= 184

```
#Connection between estimates from the PH and AFT models
#using beta_PH=kappa*beta_AFT, or beta_AFT=beta_PH/kappa
```

```
mod.weib.ph$coefficients["as.factor(treat)2"]/
exp(mod.weib.ph$coefficients["logRho"])
```

```
as.factor(treat)2
-0.1398726
```

```
mod.weib.aft$coefficients["as.factor(treat)2"]
```

```
as.factor(treat)2
0.1398726
```

- Under an AFT model we have $T_1 = T_0 e^{-\beta_{AFT}}$ where T_0 denotes event time under placebo and T_1 denotes event time under active treatment. Using the Weibull model we have

$S(t|X) = \exp\{-\lambda e^{\kappa\beta_{AFT}X} t^\kappa\}$, where $X = 0$ denotes placebo and $X = 1$ denotes active treatment. We can also write $S_1(t) = S_0(te^{\beta_{AFT}})$, where $S_0(t)$ and $S_1(t)$ are the survival functions in the placebo and active treatment groups respectively.

- The results give the estimate $\hat{\beta}_{AFT} = -0.14$. The acceleration factor is $e^{\hat{\beta}_{AFT}} = e^{-0.14} = 0.87$. The inverse of this is $e^{-\hat{\beta}_{AFT}} = e^{0.14} = 1.15$. So in this example we have $T_1 = T_0 \times 1.15$. The interpretation is that the event tends to occur later in the active treatment group, with event times in the active treatment group tending to be 1.15 times those in the control group.
- More formally $\hat{S}_1(t) = \hat{S}_0(te^{\hat{\beta}_{AFT}}) = \hat{S}_0(t \times 0.87)$. The survival curve in the treatment group at time t is what was in the placebo group at time $t \times 0.87$. The 95% CI for the acceleration factor includes the value 1, meaning there is no evidence that the distribution of event times differs in the two groups.
- We have $\beta_{PH} = \kappa\beta_{AFT}$, or $\beta_{AFT} = \beta_{PH}/\kappa$. In this example $\hat{\beta}_{PH} = -0.148$, $\hat{\kappa} = 1.059$, giving $\beta_{PH}/\kappa = -0.148/1.059 = -0.139$, which is our estimate $\hat{\beta}_{AFT}$.

Question 3

- A log-logistic model is another type of AFT model. Use `survreg` to fit a log-logistic model with treatment as the explanatory variable.
- Find the estimates of the acceleration factor, and interpret the results.
- Compare the fit of the Weibull model and the log-logistic model using the AIC. Which model provides the better fit?

```
#Log-logistic model
mod.loglog=survreg(Surv(time,d)~as.factor(treat),data=pbpc,
                   dist="loglogistic")
summary(mod.loglog)

#AICs from the Weibull and log-logistic models
AIC(mod.weib.aft)
AIC(mod.loglog)
```

SOLUTION

```
#Log-logistic model
mod.loglog=survreg(Surv(time,d)~as.factor(treat),data=pbpc,
                   dist="loglogistic")
summary(mod.loglog)
```

```
Call:
survreg(formula = Surv(time, d) ~ as.factor(treat), data = pbc,
        dist = "loglogistic")
```

	Value	Std. Error	z	p
(Intercept)	1.6719	0.1578	10.60	<2e-16
as.factor(treat)2	0.0497	0.2224	0.22	0.823
Log(scale)	-0.2197	0.0876	-2.51	0.012

```
Scale= 0.803
```

```
Log logistic distribution
```

```
Loglik(model)= -296   Loglik(intercept only)= -296
```

```
Chisq= 0.05 on 1 degrees of freedom, p= 0.82
```

```
Number of Newton-Raphson Iterations: 4
```

```
n= 184
```

```
#AICs from the Weibull and log-logistic models
AIC(mod.weib.aft)
```

```
[1] 591.0513
```

```
AIC(mod.loglog)
```

```
[1] 597.9109
```

- The results give the estimate $\hat{\beta}_{AFT} = -0.0497$. The acceleration factor is $e^{\hat{\beta}_{AFT}} = e^{-0.0497} = 0.95$. The inverse of this is $e^{-\hat{\beta}_{AFT}} = e^{0.0497} = 1.05$. So in this example we have $T_1 = T_0 \times 1.05$. The interpretation is that the event tends to occur later in the active treatment group, with event times in the active treatment group tending to be 1.05 times those in the control group. However, looking at the p-value, there is no evidence that treatment is associated with time to death.
- The AIC from the Weibull model is 591.05 and the AIC from the log-logistic model is 597.91. As the AIC from the Weibull model is lower, the Weibull model provides a slightly better fit than the log-logistic.
- Note that R and Stata give different values for the AIC. I am not sure why - the values from R are actually what we would expect based on the AIC formula in terms of the log likelihood and the number of model parameters. I suspect Stata has subtracted a constant from the log likelihood. The difference between the AIC values from the two models is the same in Stata and R, and this is what we're looking at to compare the models.

Part C: Additive hazard models

Question 1

- Fit the additive hazards model using `treat` and `bil0` as explanatory variables.
- Write down the form of the model fitted and take a look at the estimated coefficients.
- Plot the cumulative coefficients and interpret the graphical output.

```
#Fit additive hazards model
mod.aalen=aareg(Surv(time,d)~as.factor(treat)+bil0,data=pbpc)

#The coefficients are in here:
#The first column (actually the rownames here) is the times
head(mod.aalen$coefficient)

#Plot the cumulative coefficients
par(mfrow=c(1,3))
plot(mod.aalen)
```

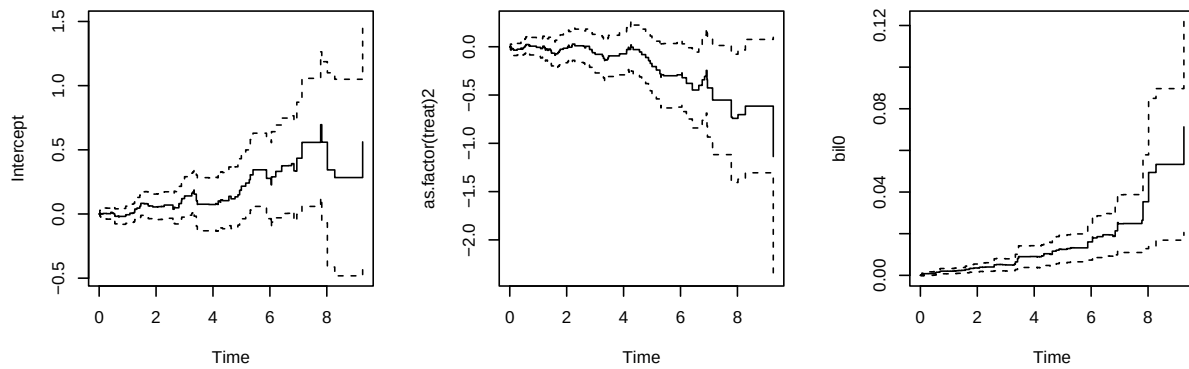
SOLUTION

```
#Fit additive hazards model
mod.aalen=aareg(Surv(time,d)~as.factor(treat)+bil0,data=pbpc)

#The coefficients are in here:
#The first column (actually the rownames here) is the times
head(mod.aalen$coefficient)
```

	Intercept	as.factor(treat)2	bil0
0.024640657	-0.004209437	0.009660735	7.746927e-05
0.024640657	0.010717828	-0.011213131	7.237878e-06
0.052019164	-0.006025511	-0.015428825	3.182515e-04
0.10403833	-0.012491216	0.006483686	2.503531e-04
0.10403833	0.011183230	-0.011426980	3.615773e-06
0.13141684	0.005004555	-0.013403327	1.301829e-04

```
#Plot the cumulative coefficients
par(mfrow=c(1,3))
plot(mod.aalen)
```

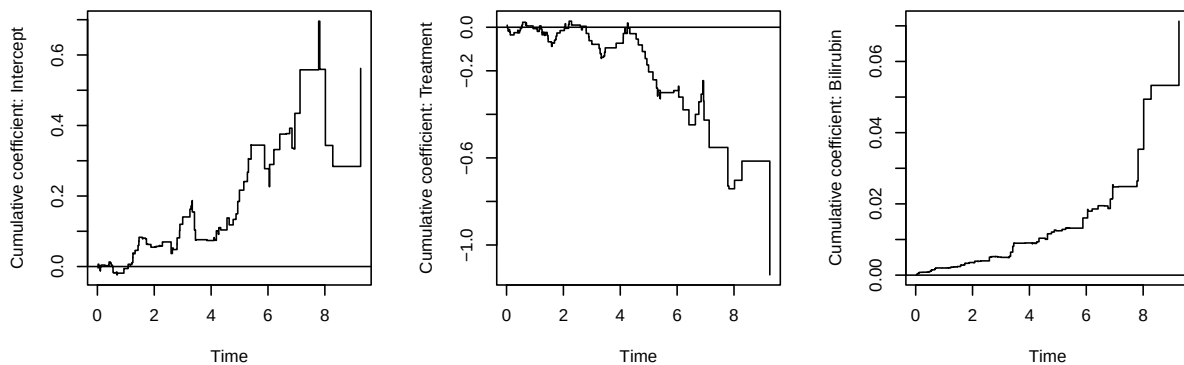


```
par(mfrow=c(1,3))

#another way of getting these plots
plot(mod.aalen$times,cumsum(mod.aalen$coefficient[,1]),type="s",
      xlab="Time",ylab="Cumulative coefficient: Intercept")
abline(h=0)

plot(mod.aalen$times,cumsum(mod.aalen$coefficient[,2]),type="s",
      xlab="Time",ylab="Cumulative coefficient: Treatment")
abline(h=0)

plot(mod.aalen$times,cumsum(mod.aalen$coefficient[,3]),type="s",
      xlab="Time",ylab="Cumulative coefficient: Bilirubin")
abline(h=0)
```



- The form of the hazard model being fitted is: $h(t|X, bil0) = \beta_0(t) + \beta_1(t)X + \beta_2(t)bil0$, where X denotes treatment as above.
- The plots show the cumulative coefficients of the additive hazards model over time. There are plots for the treatment variable and the bilirubin variable, plus a plot for the intercept.

- The plot for the intercept term is showing the estimates of the baseline cumulative hazard, i.e. the cumulative hazard in the placebo group when bilirubin is equal to 0. Overall, it is going up steadily over time. It is going up more steeply from around time 4 onwards, than in the earlier part of follow-up. There are some points in the early part of follow-up where the baseline cumulative hazard is estimated to be just less than 0 - this is something that can happen with the additive hazards model, although it is not logical that in reality that the cumulative hazard can be negative.
- The plot for treatment shows the cumulative regression coefficient fluctuating around 0 up to time 4, after which it starts to dip below zero, meaning that less hazard is being accumulated in the active treatment group. The 95% CIs for the cumulative regression coefficient always includes 0, indicating no difference in the hazards in the placebo and active treatment groups. This is conditional on bilirubin.
- The plot for bilirubin is going up more steeply over time, indicating that those with higher bilirubin accumulate increasing hazard for mortality as time goes on. This is conditional on treatment. The 95% CIs for the cumulative regression coefficient exclude 0.

Question 2

In this question we will compare survival curves estimates using the additive hazards model and using the Cox model.

The code below takes you through the following steps:

- We first need to use a different function for fitting the additive hazards model (`aalen` in the `timereg` package) because using `aareg` in the `survival` package does not allow us to automatically obtain survival estimates.
- Based on the additive hazards model fitted above, obtain the estimated survival curves for individuals 1 and 2: person 1 has `treat=1`, `bil0=30` and person 2 has `treat=2`, `bil0=30`.
- Compare the estimated survival curves with those obtained from the Cox model in Part A.

```
#Fitting the model using timereg package.
#The resample.iid=1 option is needed if we want to later use `predict`.
mod.aalen=aalen(Surv(time,d)~as.factor(treat)+bil0,data=pbcr,
                 resample.iid = 1)

#The cumulative coefficients are in here:
head(mod.aalen$cum)

#Plot the cumulative coefficients
par(mfrow=c(1,3))
```

```

plot(mod.aalen)

#Obtain survival estimates
surv.aalen<-predict(mod.aalen,
                    newdata=data.frame(treat=c(1,2),bil0=c(30,30)))
surv.person1.aalen<-surv.aalen$S0[1,]
surv.person2.aalen<-surv.aalen$S0[2,]

#Plot survival curves, and compare with those from the Cox model
par(mfrow=c(1,1))
plot(surv.aalen$time, surv.person1.aalen, type="s",lwd=2,
     xlab="Time",ylab="Survival probability")
lines(surv.aalen$time, surv.person2.aalen, type="s",lwd=2,col="grey")
lines(surv.cox$time, surv.person1.cox, type="s",lwd=2,col="black",lty=2)
lines(surv.cox$time, surv.person2.cox, type="s",lwd=2,col="grey",lty=2)
legend(x="topright",
      c("Aalen: Treat=1,bil0=30","Aalen: Treat=2,bil0=30",
        "Cox: Treat=1,bil0=30","Cox: Treat=2,bil0=30"),
      lty=c(1,1,2,2),lwd=2,
      col=c("black","grey","black","grey"),cex=0.8)

```

SOLUTION

```

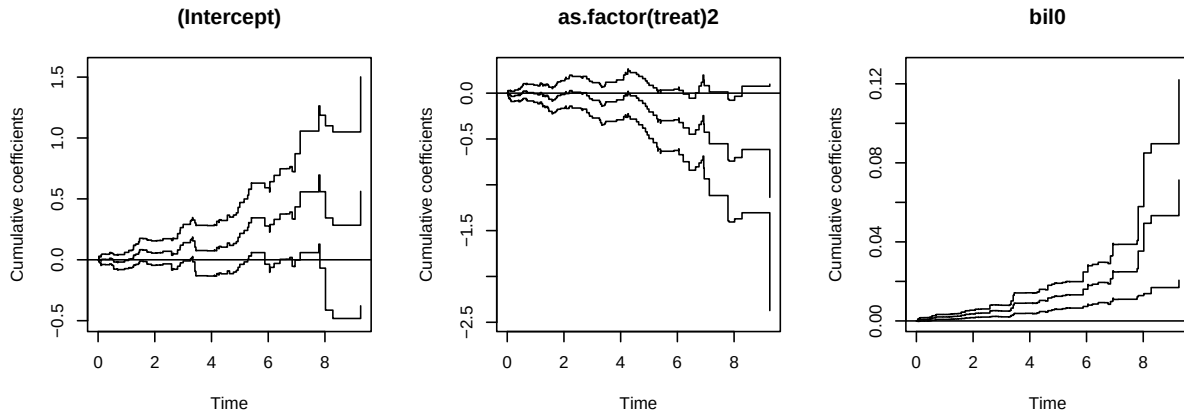
#Fitting the model using timereg package.
#The resample.iid=1 option is needed if we want to later use `predict`.
mod.aalen=aalen(Surv(time,d)~as.factor(treat)+bil0,data=pbcc,
                resample.iid = 1)

#The cumulative coefficients are in here:
head(mod.aalen$cum)

```

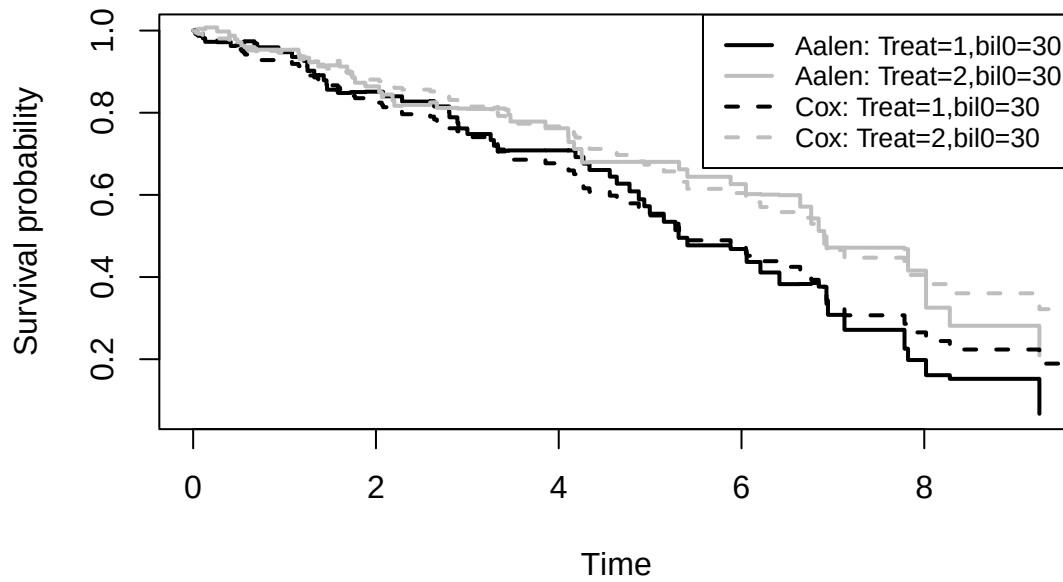
	time	(Intercept)	as.factor(treat)2	bil0
[1,]	0.00000000	0.0000000000	0.000000000	0.000000e+00
[2,]	0.02464066	-0.0042094370	0.009660735	7.746927e-05
[3,]	0.02549257	0.0065054818	-0.001545720	8.476068e-05
[4,]	0.05201916	0.0004799710	-0.016974546	4.030122e-04
[5,]	0.10403833	-0.0120112451	-0.010490860	6.533653e-04
[6,]	0.10494723	-0.0008409186	-0.021911143	6.572397e-04

```
#Plot the cumulative coefficients
par(mfrow=c(1,3))
plot(mod.aalen)
```



```
#Obtain survival estimates
surv.aalen<-predict(mod.aalen,
                    newdata=data.frame(treat=c(1,2),bil0=c(30,30)))
surv.person1.aalen<-surv.aalen$S0[1,]
surv.person2.aalen<-surv.aalen$S0[2,]

#Plot survival curves, and compare with those from the Cox model
par(mfrow=c(1,1))
plot(surv.aalen$time, surv.person1.aalen, type="s",lwd=2,
     xlab="Time",ylab="Survival probability")
lines(surv.aalen$time, surv.person2.aalen, type="s",lwd=2,col="grey")
lines(surv.cox$time, surv.person1.cox, type="s",lwd=2,col="black",lty=2)
lines(surv.cox$time, surv.person2.cox, type="s",lwd=2,col="grey",lty=2)
legend(x="topright",
      c("Aalen: Treat=1,bil0=30","Aalen: Treat=2,bil0=30",
        "Cox: Treat=1,bil0=30","Cox: Treat=2,bil0=30"),
      lty=c(1,1,2,2),lwd=2,
      col=c("black","grey","black","grey"),cex=0.8)
```



- The estimated survival curves are similar. They both have the ‘stepwise’ nature, as both are semi-parametric estimates. The additive hazards model allows the curves to move together and apart over time, as we see a bit in the middle. This is forced not to happen under the proportional hazards model.
- In this example the additive hazards model in fact gives some survival probability estimates that are slightly above 1, which can happen under this model.